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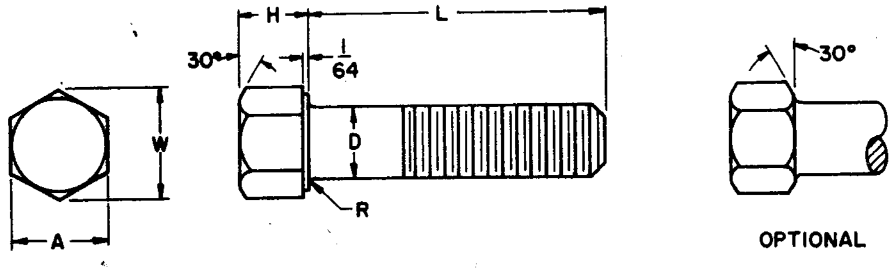

NOMINAL SIZE			1/4	5/16	3/8	7/16	1/2	9/16	5/8
THREADS PER INCH			28UNF	24UNF	24UNF	20UNF	20UNF	18UNF	18UNF
BODY DIAMETER	D	Max	.2500	.3125	.3750	.4375	.5000	.5625	.6250
		Min	.2450	.3065	.3690	.4305	.4930	.5545	.6170
WIDTH ACROSS FLATS	A	Max	.4375	.5000	.5625	.6250	.7500	.8125	.9375
		Min	.4280	.4890	.5510	.6120	.7360	.7980	.9220
WIDTH ACROSS CORNERS	W	Max	.5050	.5770	.6500	.7220	.8660	.9380	1.0330
		Min	.4880	.5570	.6280	.6980	.8400	.9100	1.0510
HEAD HEIGHT	H	Max	.1630	.2110	.2430	.2910	.3230	.3710	.4030
		Min	.1500	.1950	.2260	.2720	.3020	.3480	.3780
RADIUS	R	Max	.023	.023	.023	.023	.023	.041	.041
		Min	.009	.009	.009	.009	.009	.021	.021
LENGTH	L	Tolerance	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number
3/8		+0 -1/32	1	27	53				
7/16			2	28	54	79			
1/2			3	29	55	80	104		
9/16			4	30	56	81	105	130	
5/8			5	31	57	82	106	131	155
3/4			6	32	58	83	107	132	156
7/8			7	33	59	84	108	133	157
1			8	34	60	85	109	134	158
1-1/8		+0 -1/16	9	35	61	86	110	135	159
1-1/4			10	36	62	87	111	136	160
1-3/8			11	37	63	88	112	137	161
1-1/2			12	38	64	89	113	138	162
1-3/4			13	39	65	90	114	139	163
2			14	40	66	91	115	140	164
2-1/4		+0 -3/32	15	41	67	92	116	141	165
2-1/2			16	42	68	93	117	142	166
2-3/4			17	43	69	94	118	143	167
3			18	44	70	95	119	144	168
3-1/4			19	45	71	96	120	145	169
3-1/2			20	46	72	97	121	146	170
3-3/4			21	47	73	98	122	147	171
4			22	48	74	99	123	148	172
4-1/4			23	49	75	100	124	149	173
4-1/2			24	50	76	101	125	150	174
4-3/4			25	51	77	102	126	151	175
5			26	52	78	103	127	152	176
5-1/2							128	153	177
6							129	154	178

Ⓐ CANCELLED
SEE SHEET 2

MATERIAL: Low Carbon Steel, Grade 2 of procurement specification. Minimum tensile strengths: 69,000 psi for diameters through 1/2"; 64,000 psi for diameters 9/16" through 3/4"; 55,000 psi for diameters over 3/4".

FINISH: Plain (untreated).

THREADS: The threads shall be in accordance with Screw-Thread Standards for Federal Services Handbook H-28.

MINIMUM THREAD LENGTH: Shall be twice the diameter plus 1/4" for lengths up to and including 6". The tolerance shall be plus 3/16" or 2-1/2 threads, whichever is greater. On products that are too short for minimum thread lengths, the distance from the bearing surface of the head to the first complete thread shall not exceed the length of 2-1/2 threads, for sizes up to and including 1" and 3-1/2 threads for sizes larger than 1".

NOTES:
(1) Referenced documents of issue in effect on date of invitation for bids shall apply.
(2) In case of conflict with any referenced document, this standard will govern.
(3) The MS part number consists of the MS sheet number, plus the dash number. Example: MS 35290-1.
(4) All dimensions in inches.

Ⓐ For Changes see Sheets 1 and 2.

REVISED Ⓐ 30 APRIL 63

APPROVED 4 MAR 54

P.A. WC Other Cust SHIPS ASD	TITLE SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), LOW CARBON STEEL, PLAIN FINISH, UNF-2A	MILITARY STANDARD MS35290
PROCUREMENT SPECIFICATION FF-S-85	SUPERSEDES:	SHEET 1 OF 2

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

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NOMINAL SIZE				3/4	7/8	1	1-1/8	1-1/4	1-3/8	1-1/2
THREADS PER INCH				16UNF	14UNF	12UNF	12UNF	12UNF	12UNF	12UNF
BODY DIAMETER	D		Max Min	.7500 .7410	.8750 .8660	1.0000 0.9900	1.1250 1.1140	1.2500 1.2390	1.3750 1.3630	1.5000 1.4880
WIDTH ACROSS FLATS	A		Max Min	1.1250 1.1000	1.3125 1.2850	1.5000 1.4690	1.6875 1.6310	1.8750 1.8120	2.0625 1.9940	2.2500 2.1750
WIDTH ACROSS CORNERS	W		Max Min	1.2990 1.2540	1.5160 1.4650	1.7320 1.6750	1.9490 1.8590	2.1650 2.0660	2.3820 2.2730	2.5980 2.4800
HEAD HEIGHT	H		Max Min	.4830 .4550	.5630 .5310	.6270 .5910	.7180 .6580	.8130 .7490	.8780 .8100	.9740 .9020
RADIUS	R		Max Min	.041 .021	.062 .041	.093 .062	.093 .062	.093 .062	.093 .062	.093 .062
LENGTH L	TOLERANCE			Dash Number	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number
	Size 3/4 and under	Size 7/8 and over								
3/4	+0 -1/32	+0 -1/16		179						
7/8				180	202					
1				181	203	224				
1-1/8	+0 -1/16	+0 -1/8		182	204	225	245			
1-1/4				183	205	226	246	265		
1-3/8				184	206	227	247	266	284	
1-1/2				185	207	228	248	267	285	302
1-3/4				186	208	229	249	268	286	303
2				187	209	230	250	269	287	304
2-1/4	+0 -3/32	+0 -3/16		188	210	231	251	270	288	305
2-1/2				189	211	232	252	271	289	306
2-3/4				190	212	233	253	272	290	307
3				191	213	234	254	273	291	308
3-1/4				192	214	235	255	274	292	309
3-1/2				193	215	236	256	275	293	310
3-3/4				194	216	237	257	276	294	311
4				195	217	238	258	277	295	312
4-1/4				196	218	239	259	278	296	313
4-1/2				197	219	240	260	279	297	314
4-3/4				198	220	241	261	280	298	315
5				199	221	242	262	281	299	316
5-1/2				200	222	243	263	282	300	317
6				201	223	244	264	283	301	318

NOTE: For illustration, material, finish and other pertinent data, see sheet 1.

Ⓐ CANCELLED AFTER 30 April 1963. USE MS90726. INTERCHANGEABILITY RELATIONSHIP - MS90726 can replace the cancelled screws of the same size. The cancelled MS35290 cannot always replace MS90726 and should be used until existing stock is depleted. Then use only the superseding MS screws of the same size for design and replacement.

Ⓐ CANCELLED
SEE NOTE

APPROVED 4 MAR 54 REVISED Ⓐ FOR CHANGES SEE SHEETS 1 AND 2.

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P.A.
Other Cust
ASD

TITLE
SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT),
LOW CARBON STEEL, PLAIN FINISH, UNF-2A

MILITARY STANDARD
MS35290

SHEET 2 OF

PROCUREMENT SPECIFICATION
FF-S-85

SUPERSEDING:

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